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Robust differential game guidance laws design for uncertain interceptor-target engagement via adaptive dynamic programming

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ABSTRACT

In this paper, the problem of intercepting a maneuvering target is formulated as a two-player zero-sum differential game framework affected by matched uncertainties. By introducing an appropriate cost function that reflects the uncertainties, the robust control is transformed into a two-player zero-sum differential game control problem and therefore ensures the compensation of the matched uncertainties. Additionally, the corresponding Hamilton–Jacobi–Isaacs (HJI) equation is solved by constructing a critic neural network (NN). The closed-loop system and the critic NN weight estimation error are proved to be uniform ultimate boundedness (UUB) by utilising Lyapunov approach. Finally, the effectiveness of the proposed robust guidance law is demonstrated by using a nonlinear two-dimensional kinematics, assuming first-order dynamics for the interceptor and the target.

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1. Introduction

In the modern warfare, the main objective of guided missiles is to intercept the targets successfully while optimising the energy usage to increase their range. Recently, variants of prospering missile guidance technologies have been developed to improve the guidance performance, such as proportional navigation guidance (PNG) (Ratnoo & Ghose, 2008), augmented proportional navigation guidance (APNG) (Ghosh, Ghose, & Raha, 2014), sliding-mode control (SMC) guidance (Kumar, Rao, & Ghose, 2012), and linear quadratic differential game (LQDG) guidance (Shaferman & Shima, 2008). Among these techniques, PNG is the most popular and used widely in the practical missile interception engagements. However, in the presence of guidance parameter's uncertainties and external disturbances such as target maneuvers, the PNG is not able to perform well. The APNG was developed on the assumption that a target's acceleration is known to the interceptor. Nevertheless, any intelligent target, that may employ strategies in feedback form, selects its strategy independently and it is hard to predict its future course of strategies. The differential game theory, especially pursuit-evasion games, provides a framework on which the analysis of the conflict situation between two independent selfish adversaries is required (Rajarshi & Debasish, 2015). Hereafter, LQDG was applied successfully in target interception problems, as well as the state-dependent Riccati equation (SDRE)-based differential game guidance (Bardhan & Ghose,

2012). But both LQDG- and SDRE-based guidance laws are either derived from linearised geometry around the collision course that needs estimation of time-to-go or with no consideration of the robustness to internal or external disturbances. The SMC theory provides a robust method that can handle nonlinearity in system states, large modelling errors, and uncertainties. The guidance approach based on SMC was studied in the past few years (Kumar et al., 2012; Kumar, Rao, & Ghose, 2014), whereas most of the guidance laws are discontinuous that may demand high rates of control and may lead to chattering during implementing these guidance laws.

In order to design a robust guidance law with respect to uncertainties for interceptor–target engagement dynamics, as well as relax the need for information about the target's maneuver, the H_∞ guidance law (Yang & Chen, 1998) and the SDRE guidance law (Bardhan & Ghose, 2015; Rajarshi & Debasish, 2015) were proposed with nonlinear kinematics by solving the corresponding Hamilton–Jacobi–Isaacs (HJI) equation. However, it is usually intractable or impossible to find its Nash solution by analytical methods for nonlinear differential game systems, which limits the application of the guidance method. Solution methods are generally offline.

Adaptive dynamic programming (ADP) as an effective intelligent control method has played an important role in seeking solutions for the optimal control and differential games (Zhang, Zhang, Luo, & Yang, 2013). Reinforcement learning (RL), as the synonym of the ADP, has

been used widely to cope with the nonlinear optimal control and differential game problem. The general structure, named actor-critic architecture, is used to implement the ADP/RL, where the critic is a network which approximates the cost function while the actor is another network which approximates the control policy. In recent years, various related works have been performed by using the ADP/RL technique, such as stabilising control problem (Vamvoudakis & Lewis, 2010; Vamvoudakis, Vrabie, & Lewis, 2011; Vrabie & Lewis, 2009), tracking control problem (Gao & Jiang, 2015; Kamalapurkar, Dinh, Bhasin, & Dixon, 2015; Liu & Tong, 2015), output feedback control problem (Zhu, Modares, Peen, Lewis, & Baozeng, 2015), constrained-input control problem (Liu, Yang, Wang, & Wei, 2015; Xiong, Derong, & Yuzhu, 2013), H_∞ control problem (Modares, Lewis, & Sistani, 2014), differential game problem (Vamvoudakis, Vrabie, & Lewis, 2012; Yasini, Sistani, & Karimpour, 2015), and robust control problem (Fu, Fu, & Chai, 2015; Wang, Liu, Li, & Ma, 2014; Jiang and Jiang, 2014).

In practical interceptor–target engagement, the external disturbances and modelling errors can severely degrade the closed-loop system performance. A possible solution is the adoption of the robust strategies to improve the guidance performance and to decrease the zero effort miss distance (ZEMD). Unfortunately, there are few results on robust optimal control of the uncertain nonlinear systems based on ADP, not to mention the interceptor–target engagement differential game problem. In Fu et al. (2015), the authors develop a robust differential game controller for linear systems with matched uncertainties by expending the results of Zhang and Zhang (2014). Although the proposed robust controller is implemented without any knowledge of the systems, the initial stabilising control is required, which is a rather restrictive condition for the nonlinear interceptor–target engagement dynamics.

In this paper, expending the results of Wang et al. (2014), a robust differential game control approach for a class of nonlinear systems with uncertainties is established by modifying the Nash solution of the nominal system under an appropriate cost function. The ADP technique that does not suffer from the curse of dimensionality is utilised to solve the corresponding HJI equation by constructing a critic network. Besides, inspired by the work of Dierks and Jagannathan (2010), Liu et al. (2015) and Wang et al. (2014), an additional stabilising term is introduced to ensure the system states remain bounded during the online learning process. By utilising Lyapunov's direct method, the closed-loop differential game system and the critic NN weight estimation error are proved to be uniform ultimate boundedness (UUB) and the ultimate bound of the system states is given.

The main contributions of this paper include the following:

- (1) The robust guidance law for interceptor–target engagement is investigated by utilising the differential game approach based on ADP/RL to improve the guidance performance.
- (2) In comparison with Vamvoudakis et al. (2012) and Yasini et al. (2015), a clear advantage of the proposed algorithm in this paper is that a simpler algorithm architecture is constructed, that is, only one critic network is employed instead of the two or three networks to implement the algorithm, which reduces the complexity of the computation.

The rest of this paper is organised as follows. In Section 2, we present the general problem statement and preliminaries. The robust controller of the uncertain nonlinear two-player zero-sum differential games is designed with stability proof in Section 3. Then, the ADP-based robust differential game controller is developed with stability proof in Section 4. In Section 5, the mathematical model of interceptor–target engagement used for the guidance law's derivation and simulation are presented. Finally, performance of the guidance law has been demonstrated and analysed in Section 6, followed by concluding remarks in Section 7.

Throughout this paper, R^n and $R^{n \times m}$ denote an n -dimensional Euclidean space and the set of all $n \times m$ real matrices, respectively; Ω is a compact set of R^n ; when ξ is a vector, $\|\xi\|$ means the Euclidean norm of ξ ; when A is a matrix, $\|A\|$ denotes the two norms of A ; the transpose of the matrix M is presented by M^T ; $P > 0$ denotes the matrix P is positive-definite; x^* denotes the optimal value of variable x .

2. Problem statement and preliminaries

In this section, we study the continuous-time uncertain nonlinear two-player zero-sum differential games given by

$$\begin{aligned} \dot{x}(t) = & f(x(t)) + g(x(t))(\bar{u}(t) + \bar{d}(x(t))) \\ & + k(x(t))(\bar{v}(t) + \bar{b}(x(t))) \end{aligned} \quad (1)$$

where $x(t) \in R^n$ is the state vector, $\bar{u}(t) \in R^k$ and $\bar{v}(t) \in R^m$ are the two control input vectors for two players, $f(\cdot)$, $g(\cdot)$ and $k(\cdot)$ are differentiable in their arguments with $f(0) = 0$, and $\bar{d}(x(t))$, $\bar{b}(x(t))$ are the unknown nonlinear-matched uncertainties for the players. Assume $f(x)$ is local Lipschitz, such that system (1) is controllable. Let $x(0) = x_0$ be the initial state. Assume that

$\bar{d}(0) = 0, \bar{b}(0) = 0$, so that $x = 0$ is an equilibrium point of system (1).

The nominal system (i.e. system (1) without uncertainties) is described by

$$\dot{x}(t) = f(x(t)) + g(x(t))u(t) + k(x(t))v(t) \quad (2)$$

For convenience, denote $x(t)$ as x , in which time variable t is omitted in some subsequent formulas.

For system (1), in order to deal with the matched uncertainties, the control pair $\bar{u}(t)$ and $\bar{v}(t)$ should be found, such that the closed-loop system is globally asymptotically stable for all uncertainties $\bar{d}(x)$ and $\bar{b}(x)$. To this end, in this paper, we convert this problem into finding Nash solution for corresponding nominal zero-sum differential games with appropriate cost function.

Let $R_1 \in R^{k \times k}$, $R_2 \in R^{m \times m}$ be two symmetric positive definite matrices. Then, we assume that $d(x) = \frac{1}{R_1} \bar{d}(x)$, $b(x) = \frac{1}{R_2} \bar{b}(x)$, where $d(x) \in R^k$ and $b(x) \in R^m$ are both bounded by the known function $d_M(x)$ and $b_M(x)$, i.e. $\|d(x)\| \leq d_M(x)$ and $\|b(x)\| \leq b_M(x)$ with $d_M(0) = b_M(0) = 0$.

For nominal system (2), considering the uncertainties corresponding to system (1), the objective is to design one control input $u(t)$ minimising the infinite-time cost function and another control input $v(t)$ maximising the infinite-time cost function given by

$$J(x_0) = \int_0^\infty \{Q(x) + u^T(t)R_1u(t) - v^T(t)R_2v(t)\} dt \quad (3)$$

with $Q(x) = d_M^2(x) + cb_M^2(x) + x^T Q_1 x$, Q_1 is a symmetric positive definite matrix, the positive scalar $c > 1$ is indispensable and it will be a crucial design parameter that impacts the convergence of the system in this paper.

Remark 2.1: The term $Q(x)$ that contains several parts appropriately reflects the uncertainties and regulation simultaneously. Clearly, it is positive. According to the Nash–Pontryagin Minimax principle, not only the regulation error but also the uncertainties are minimised by the designed controller, which indicates the robustness to the uncertainties. The cost function described in (3) gives a modification with respect to the zero-sum differential game problem without uncertainties.

The problem of interest in this paper is to find an optimal control pair $u(t)$ and $v(t)$ such that the dynamic system (2) is transferred from the given initial state x_0 to an arbitrary final state and such that the cost function (3) is minimised with respect to $u(t)$ and maximised with respect to $v(t)$.

Define the Hamiltonian function of system (2) as follows:

$$H(x, u, \nabla J(x)) = Q(x) + u^T(t)R_1u(t) - v^T(t)R_2v(t) + (\nabla J(x))^T (f(x) + g(x)u(t) + k(x)v(t)) \quad (4)$$

with $J(0) = 0$, and the symbol $\nabla(\cdot) = \partial(\cdot)/\partial x$ is the notation of gradient operation, such that $\nabla J(x) = \partial J(x)/\partial x$.

In the Nash–Pontryagin Minimax principle, we have expressed the necessary condition for the Hamiltonian function $H(\cdot)$ to have a Nash equilibrium or a special type of saddle point with respect to the control pair $(u(t), v(t))$ at the optimal control pair $(u^*(t), v^*(t))$ by the two inequalities

$$H(x^*, u^*(t), v(t), \nabla J^*(x)) \leq H(x^*, u^*(t), v^*(t), \nabla J^*(x)) \leq H(x^*, u(t), v^*(t), \nabla J^*(x)) \quad (5)$$

According to the HJI theory, the optimal cost function $J^*(x)$ is given by

$$J^*(x) = \max_{v(t)} \min_{u(t)} \int_0^\infty \{Q(x) + u^T(t)R_1u(t) - v^T(t)R_2v(t)\} dt \quad (6)$$

with $J^*(0) = 0$. Then, the optimal cost function $J^*(x)$ can be obtained by solving the HJI equation

$$0 = \max_{v(t)} \min_{u(t)} H(x, u(t), v(t), \nabla J^*(x)) \quad (7)$$

Assume the saddle point of (7) exists and is unique. Then, the optimal control pair is

$$\begin{aligned} u^*(t) &= -\frac{1}{2}R_1^{-1}g^T(x)\nabla J^*(x) \\ v^*(t) &= \frac{1}{2}R_2^{-1}k^T(x)\nabla J^*(x) \end{aligned} \quad (8)$$

Based on (4) and (8), the HJI equation (7) becomes

$$\begin{aligned} 0 &= Q(x) - \frac{1}{4}(\nabla J^*(x))^T g(x)R_1^{-1}g^T(x)\nabla J^*(x) \\ &\quad + \frac{1}{4}(\nabla J^*(x))^T k(x)R_2^{-1}k^T(x)\nabla J^*(x) + (\nabla J^*(x))^T f(x) \end{aligned} \quad (9)$$

with $J^*(0) = 0$.

3. Robust controller design of uncertain nonlinear two-player zero-sum differential games

In this section, inspired by the work of Wang et al. (2014), considering system (1) with uncertainties, in order to compensate the matched uncertainties, we modify the optimal control pair (8) of system (2) as follows:

$$\begin{aligned}\bar{u}(t) &= -\frac{\varepsilon_1}{2} R_1^{-1} g^T(x) \nabla J^*(x) \\ \bar{v}(t) &= \frac{1}{2\varepsilon_2} R_2^{-1} k^T(x) \nabla J^*(x)\end{aligned}\quad (10)$$

in which $\varepsilon_1 > 1$ and $\varepsilon_2 > 1$ are feedback gains.

Now, we present the following lemma to indicate that the modified control pair has infinite gain margin.

Lemma 3.1: *For nominal system (2), the modified control pair given in (10) ensures that the closed-loop system is asymptotically stable as long as the feedback gains satisfy $\varepsilon_1 \geq \frac{1}{2}$, $\varepsilon_2 \geq 2$.*

Proof: We denote the optimal cost function $J^*(x)$ is a Lyapunov function. In the sight of (3), we can easily find that $J^*(x)$ is positive definite. Considering (9) and (10), the derivative of $J^*(x)$ along the trajectory of the closed-loop system is

$$\begin{aligned}\dot{J}^*(x) &= (\nabla J^*(x))^T (f(x) + g(x)\bar{u}(t) + k(x)\bar{v}(t)) \\ &= -Q(x) + \frac{1}{4} (\nabla J^*(x))^T g(x) R_1^{-1} g^T(x) \nabla J^*(x) \\ &\quad - \frac{1}{4} (\nabla J^*(x))^T k(x) R_2^{-1} k^T(x) \nabla J^*(x) \\ &\quad - \frac{\varepsilon_1}{2} (\nabla J^*(x))^T g(x) R_1^{-1} g^T(x) \nabla J^*(x) \\ &\quad + \frac{1}{2\varepsilon_2} (\nabla J^*(x))^T k(x) R_2^{-1} k^T(x) \nabla J^*(x) \\ &= -Q(x) - \frac{1}{2} \left(\varepsilon_1 - \frac{1}{2} \right) \left\| R_1^{-\frac{1}{2}} g^T(x) \nabla J^*(x) \right\|^2 \\ &\quad - \frac{1}{2} \left(\frac{1}{2} - \frac{1}{\varepsilon_2} \right) \left\| R_2^{-\frac{1}{2}} k^T(x) \nabla J^*(x) \right\|^2\end{aligned}\quad (11)$$

Hence, $J^*(x) < 0$, whenever $\varepsilon_1 \geq \frac{1}{2}$, $\varepsilon_2 \geq 2$ and $x \neq 0$. Then, the conditions for Lyapunov local stability theory are satisfied. ■

Theorem 3.1: *For system (1), there always exist two feedback gains satisfying $\varepsilon_1 \geq 1$ and $\varepsilon_2 \geq \frac{2c}{c-1}$, where $c > 1$ is designed in (3), such that the modified control pair developed in (10) ensures that the closed-loop system is asymptotically stable.*

Proof: We select $\bar{L}(x) = J^*(x)$ as the Lyapunov function candidate. Taking the time derivative of $\bar{L}(x)$ along the

trajectory of the closed-loop system, we obtain

$$\begin{aligned}\dot{\bar{L}}(x) &= (\nabla J^*(x))^T (f(x) + g(x)(\bar{u}(t) + \bar{d}(x)) \\ &\quad + k(x)(\bar{v}(t) + \bar{b}(x)))\end{aligned}\quad (12)$$

Based on (9) and (10), we find that

$$\begin{aligned}\dot{\bar{L}}(x) &= -Q(x) - \frac{1}{2} \left(\varepsilon_1 - \frac{1}{2} \right) \left\| (\nabla J^*(x))^T g(x) R_1^{-\frac{1}{2}} \right\|^2 \\ &\quad - \frac{1}{2} \left(\frac{1}{2} - \frac{1}{\varepsilon_2} \right) \left\| (\nabla J^*(x))^T k(x) R_2^{-\frac{1}{2}} \right\|^2 \\ &\quad + (\nabla J^*(x))^T g(x) \bar{d}(x) + (\nabla J^*(x))^T k(x) \bar{b}(x) \\ &\leq -x^T Q_1 x - \left\{ \frac{1}{2} \left(\varepsilon_1 - \frac{1}{2} \right) \left\| (\nabla J^*(x))^T g(x) R_1^{-\frac{1}{2}} \right\|^2 \right. \\ &\quad + \frac{1}{2} \left(\frac{1}{2} - \frac{1}{\varepsilon_2} \right) \left\| (\nabla J^*(x))^T k(x) R_2^{-\frac{1}{2}} \right\|^2 \\ &\quad - \left\| (\nabla J^*(x))^T g(x) R_1^{-\frac{1}{2}} \right\| d_M(x) \\ &\quad \left. - \left\| (\nabla J^*(x))^T k(x) R_2^{-\frac{1}{2}} \right\| b_M(x) + d_M^2(x) + cb_M^2(x) \right\}\end{aligned}\quad (13)$$

Denote $\xi = [d_M(x), b_M(x), \|(\nabla J^*(x))^T g(x) R_1^{-\frac{1}{2}}\|, \|(\nabla J^*(x))^T k(x) R_2^{-\frac{1}{2}}\|]^T$. ■

Then, we have $\dot{\bar{L}}(x) \leq -x^T Q_1 x - \xi^T \Theta \xi$, where

$$\Theta = \begin{bmatrix} 1 & 0 & -\frac{1}{2} & 0 \\ 0 & c & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 & \frac{1}{2} \left(\varepsilon_1 - \frac{1}{2} \right) & 0 \\ 0 & -\frac{1}{2} & 0 & \frac{1}{2} \left(\frac{1}{2} - \frac{1}{\varepsilon_2} \right) \end{bmatrix}\quad (14)$$

From (14), if we design $\varepsilon_1 \geq 1$, $\varepsilon_2 \geq \frac{2c}{c-1}$, then $\Theta \geq 0$, and thus we guarantee $\dot{\bar{L}}(x) < 0$, which implies that the closed-loop system is asymptotically stable.

According to Theorem 3.1, the modified control pair $\bar{u}(t)$ and $\bar{v}(t)$ achieves robustness to the system (1) affected by uncertainties. Next, we will prove the property of optimality possessed by the modified control pair.

For system (1), we define the following cost function:

$$\bar{J}(x_0) = \int_0^\infty \left\{ \bar{Q}(x) + \frac{1}{\varepsilon_1} \bar{u}(t) R_1 \bar{u}(t) - \varepsilon_2 \bar{v}(t) R_2 \bar{v}(t) \right\} dt\quad (15)$$

where

$$\begin{aligned}\bar{Q}(x) &= d_M^2(x) + cb_M^2(x) + x^T Q_1 x - (\nabla J^*(x))^T g(x) \bar{d}(x) \\ &\quad - (\nabla J^*(x))^T k(x) \bar{b}(x) + \frac{1}{4} (\varepsilon_1 - 1)\end{aligned}$$

$$\begin{aligned}
& (\nabla J^*(x))^T g(x) R_1^{-1} g^T(x) \nabla J^*(x) - \frac{1}{4} \left(1 - \frac{1}{\varepsilon_2}\right) \\
& (\nabla J^*(x))^T k(x) R_2^{-1} k^T(x) \nabla J^*(x)
\end{aligned} \quad (16)$$

Lemma 3.2: *There exist two feedback gains satisfying $\varepsilon_1 \geq 2$, $\varepsilon_2 \geq \frac{c}{c-1}$, where $c > 1$ is designed in (3), such that the function $\bar{Q}(x)$ is positive definite.*

Proof: Adding and subtracting $\frac{1}{\varepsilon_1-1} d^T(x) d(x)$ and $\frac{\varepsilon_2}{\varepsilon_2-1} b^T(x) b(x)$ to the right hand of (16), and by means of some simple mathematical transformation, one can find

$$\begin{aligned}
\bar{Q}(x) = & Q(x) - (\nabla J^*(x))^T g(x) \bar{d}(x) - (\nabla J^*(x))^T k(x) \bar{b}(x) \\
& + \frac{1}{4(\varepsilon_1-1)} [(\varepsilon_1-1)^2 (\nabla J^*(x))^T \\
& \times g(x) R_1^{-1} g^T(x) \nabla J^*(x) + 4d^T(x) d(x)] \\
& + \frac{\varepsilon_2}{4(\varepsilon_2-1)} \left[\left(1 - \frac{1}{\varepsilon_2}\right)^2 (\nabla J^*(x))^T \right. \\
& \times k(x) R_2^{-1} k^T(x) \nabla J^*(x) + 4b^T(x) b(x) \left. \right] \\
& - \frac{1}{\varepsilon_1-1} d^T(x) d(x) - \frac{\varepsilon_2}{\varepsilon_2-1} b^T(x) b(x)
\end{aligned} \quad (17)$$

Applying $d(x) = R_1^{-\frac{1}{2}} \bar{d}(x)$, $b(x) = R_2^{-\frac{1}{2}} \bar{b}(x)$ and $Q(x) = d_M^2(x) + cb_M^2(x) + x^T Q_1 x$ yields

$$\begin{aligned}
\bar{Q}(x) = & d_M^2(x) + cb_M^2(x) + x^T Q_1 x \\
& - \frac{1}{\varepsilon_1-1} d^T(x) d(x) - \frac{\varepsilon_2}{\varepsilon_2-1} b^T(x) b(x) \\
& + \frac{1}{4(\varepsilon_1-1)} [(\varepsilon_1-1) (\nabla J^*(x))^T g(x) R_1^{-\frac{1}{2}} - 2d^T(x)] \cdot \\
& \times [(\varepsilon_1-1) (\nabla J^*(x))^T g(x) R_1^{-\frac{1}{2}} - 2d^T(x)]^T \\
& + \frac{\varepsilon_2}{4(\varepsilon_2-1)} \left[\left(1 - \frac{1}{\varepsilon_2}\right) (\nabla J^*(x))^T k(x) R_2^{-\frac{1}{2}} - 2b^T(x) \right] \cdot \\
& \times \left[\left(1 - \frac{1}{\varepsilon_2}\right) (\nabla J^*(x))^T k(x) R_2^{-\frac{1}{2}} - 2b^T(x) \right]^T
\end{aligned} \quad (18)$$

Recalling $\|d(x)\| \leq d_M(x)$, $\|b(x)\| \leq b_M(x)$ and the conditions $\varepsilon_1 \geq 2$, $\varepsilon_2 \geq \frac{c}{c-1}$, we can obtain

$$\begin{aligned}
\bar{Q}(x) \geq & x^T Q_1 x + d_M^2(x) + cb_M^2(x) \\
& - \frac{1}{\varepsilon_1-1} d^T(x) d(x) - \frac{\varepsilon_2}{\varepsilon_2-1} b^T(x) b(x) \\
\geq & x^T Q_1 x + \frac{\varepsilon_1-2}{\varepsilon_1-1} d_M^2(x) \\
& + \frac{(c-1)\varepsilon_2-c}{\varepsilon_2-1} b_M^2(x) \geq 0
\end{aligned} \quad (19)$$

This proves that $Q(x)$ is a positive definite function.

Theorem 3.2: *Consider system (1) with infinite-time cost function (15). There always exist two feedback gains $\varepsilon_1 \geq 2$ and $\varepsilon_2 \geq \frac{2c}{c-1}$, such that the modified control pair obtained by (10) is the asymptotically stabilising Nash solution of the differential game problem.*

Proof: The Hamiltonian function of system (1) with infinite-time cost function (15) is

$$\begin{aligned}
H(x, \bar{u}(t), \bar{v}(t), \nabla \bar{J}(x)) \\
= & \bar{Q}(x) + \frac{1}{\varepsilon_1} \bar{u}(t) R_1 u(t) - \varepsilon_2 \bar{v}(t) R_2 v(t) \\
& + (\nabla \bar{J}(x))^T (f(x) + g(x)(\bar{u}(t) + \bar{d}(x)) \\
& + k(x)(\bar{v}(t) + \bar{b}(x)))
\end{aligned} \quad (20)$$

Replacing $\bar{J}(x)$ with $J^*(x)$ and considering (16), Equation (20) becomes

$$\begin{aligned}
H(x^*, \bar{u}^*(t), \bar{v}^*(t), \nabla J^*(x)) \\
= & d_M^2(x) + cb_M^2(x) + x^T Q_1 x + (\nabla J^*(x))^T f(x) \\
& + (\nabla J^*(x))^T g(x) \bar{u}(t) + (\nabla J^*(x))^T k(x) \bar{v}(t) \\
& + \frac{1}{4}(\varepsilon_1-1) (\nabla J^*(x))^T g(x) R_1^{-1} g^T(x) \nabla J^*(x) \\
& - \frac{1}{4} \left(1 - \frac{1}{\varepsilon_2}\right) (\nabla J^*(x))^T k(x) R_2^{-1} k^T(x) \nabla J^*(x) \\
& + \frac{1}{\varepsilon_1} \bar{u}(t) R_1 \bar{u}(t) - \varepsilon_2 \bar{v}(t) R_2 \bar{v}(t) + (\nabla J^*(x))^T f(x)
\end{aligned} \quad (21)$$

Using (9) and (10), we can further obtain that $H(x^*, \bar{u}^*(t), \bar{v}^*(t), \nabla J^*(x)) = 0$, which shows that $J^*(x)$ is a solution of the HJI equation of system (1). Then, we say that the control pair (10) achieves optimality with infinite-time cost function (15). That is to say, the modified control pair (10) is the robust optimal control of the nonlinear two-player zero-sum differential game system (1) as long as the given feedback gains satisfy

$$\varepsilon_1 \geq 2, \quad \varepsilon_2 \geq \frac{2c}{c-1}.$$

In order to apply the critic network to approximate the Nash solution and guarantee its stability, the following common assumption is presented (Dierks & Jagannathan, 2010).

Assumption 3.1: *Let $J_s(x)$ be a continuously differentiable Lyapunov function candidate satisfying*

$$\begin{aligned}
\dot{J}_s(x) &= (\nabla J_s(x))^T \dot{x} \\
&= (\nabla J_s(x))^T (f(x) + g(x)u^*(t) + k(x)v^*(t)) < 0
\end{aligned} \quad (22)$$

Assume that there exists a positive definite matrix $\Lambda(x)$ such that the following relation holds:

$$\begin{aligned}
& (\nabla J_s(x))^T (f(x) + g(x)u^*(t) + k(x)v^*(t)) \\
&= -(\nabla J_s(x))^T \Lambda(x) \nabla J_s(x)
\end{aligned} \quad (23)$$

Let $\lambda_{\min}$ and $\lambda_{\max}$ be the minimum and maximum eigenvalues of matrix $\Lambda(x)$, then the inequality $\lambda_{\min} \|\nabla J_s(x)\|^2 \leq (\nabla J_s(x))^T \Lambda(x) \nabla J_s(x) \leq \lambda_{\max} \|\nabla J_s(x)\|^2$ holds.

4. The critic network via ADP and stability analysis

According to the Weierstrass higher order approximation theorem (Abu-Khalaf & Lewis, 2005), the optimal cost function can be approximated by neural network such that

$$J^*(x) = w_c^T \sigma_c(x) + \varepsilon_c(x) \quad (24)$$

where $w_c \in R^N$ is the ideal weight, $\sigma_c(x) \in R^N$ is called the neural network activation function vector, N is the number of neurons in the hidden layer, and $\varepsilon_c(x)$ is the neural network approximation error. Then, its derivation with respect to x is

$$\nabla J^*(x) = (\nabla \sigma_c(x))^T w_c + \nabla \varepsilon_c(x) \quad (25)$$

Substituting (25) into the optimal control pair (8) yields

$$\begin{aligned} u^*(t) &= -\frac{1}{2} R_1^{-1} g^T(x) ((\nabla \sigma_c(x))^T w_c + \nabla \varepsilon_c(x)) \\ v^*(t) &= \frac{1}{2} R_2^{-1} k^T(x) ((\nabla \sigma_c(x))^T w_c + \nabla \varepsilon_c(x)) \end{aligned} \quad (26)$$

The Hamiltonian function (4) becomes

$$\begin{aligned} 0 &= Q(x) + u^T(t) R_1 u(t) - v^T(t) R_2 v(t) \\ &\quad + ((\nabla \sigma_c(x))^T w_c + \nabla \varepsilon_c(x))^T \\ &\quad \times (f(x) + g(x)u(t) + k(x)v(t)) \end{aligned} \quad (27)$$

Since the ideal weights of the neural network, which provide the best approximate solution for (27), are unknown, the output of the critic neural network is

$$\hat{J}(x) = \hat{w}_c^T \sigma_c(x) \quad (28)$$

where $\hat{w}_c$ is the current estimated value of w_c . Similarly, we have

$$\nabla \hat{J}(x) = (\nabla \sigma_c(x))^T \hat{w}_c \quad (29)$$

According to (8) and (29), the approximate control pair can be given as

$$\begin{aligned} \hat{u}(t) &= -\frac{1}{2} R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T \hat{w}_c \\ \hat{v}(t) &= \frac{1}{2} R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T \hat{w}_c \end{aligned} \quad (30)$$

Substituting (26) into the Hamiltonian function (4) yields

$$\begin{aligned} H(x, w_c, u, v) &= Q(x) + w_c^T \nabla \sigma_c(x) f(x) \\ &\quad - \frac{1}{4} w_c^T \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T w_c \\ &\quad + \frac{1}{4} w_c^T \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T w_c + \varepsilon_H \end{aligned} \quad (31)$$

where the residual error due to the function approximation error is

$$\begin{aligned} \varepsilon_H &= (\nabla \varepsilon_c(x))^T f(x) - \frac{1}{4} (\nabla \varepsilon_c(x))^T g(x) R_1^{-1} g^T(x) \nabla \varepsilon_c(x) \\ &\quad + \frac{1}{4} (\nabla \varepsilon_c(x))^T k(x) R_2^{-1} k^T(x) \nabla \varepsilon_c(x) \\ &\quad - \frac{1}{2} (\nabla \varepsilon_c(x))^T g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T w_c \\ &\quad + \frac{1}{2} (\nabla \varepsilon_c(x))^T k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T w_c \end{aligned} \quad (32)$$

Using the approximated control pair and the estimated weight vector, the approximated Hamiltonian function can be derived as

$$\begin{aligned} H(x, \hat{w}_c, \hat{u}, \hat{v}) &= Q(x) + \hat{w}_c^T \nabla \sigma_c(x) f(x) \\ &\quad - \frac{1}{4} \hat{w}_c^T \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T \hat{w}_c \\ &\quad + \frac{1}{4} \hat{w}_c^T \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T \hat{w}_c \triangleq e_c \end{aligned} \quad (33)$$

with e_c a residual equation error.

Define the critic weight estimation error $\tilde{w}_c = w_c - \hat{w}_c$, then, based on (31) and (33), the residual equation error can be revised as

$$\begin{aligned} e_c &= -\tilde{w}_c^T \nabla \sigma_c(x) f(x) \\ &\quad - \frac{1}{4} \tilde{w}_c^T \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T \tilde{w}_c \\ &\quad + \frac{1}{2} \tilde{w}_c^T \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T w_c \\ &\quad + \frac{1}{4} \tilde{w}_c^T \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T \tilde{w}_c \\ &\quad - \frac{1}{2} \tilde{w}_c^T \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T w_c - \varepsilon_H \end{aligned} \quad (34)$$

Given any feedback control pair $u(t)$ and $v(t)$, it is desired to design $\hat{w}_c$ to minimise the squared residual error

$$E_c = \frac{1}{2} e_c^T e_c \quad (35)$$

Then, $\hat{w}_c \rightarrow w_c$ and $e_c \rightarrow \varepsilon_H$.

Remark 4.1: Observing the approximate Hamiltonian function (33), the estimated weight $\hat{w}_c$ should be

designed to minimise (33). However, it does not ensure the stability of system (2) during the online learning process. Therefore, an improved tuning law should be considered to minimise (33) while considering the stability of system (2).

Select the tuning law for the critic estimated weights as the normalised gradient descent algorithm with an additional terms introduced to ensure the boundedness of system state, i.e.

$$\begin{aligned}\dot{\hat{w}}_c = & -\alpha_c \left(\frac{\partial E_c}{\partial \hat{w}_c} \right) \\ & + \frac{1}{2} \alpha_s \sum (x, \hat{u}, \hat{v}) \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) \nabla J_s(x) \\ & - \frac{1}{2} \alpha_s \sum (x, \hat{u}(t), \hat{v}(t)) \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) \nabla J_s(x)\end{aligned}\quad (36)$$

Where $\alpha_c > 0$ is the learning rate of the critic network, $\alpha_s > 0$ is the regulation parameter of the additional term. $J_s(x)$ is the Lyapunov function candidate given in Assumption 3.1, and $\Sigma(x, \hat{u}, \hat{v})$ is the additional stabilising term defined as

$$\Sigma(x, \hat{u}(t), \hat{v}(t)) = \begin{cases} 0, & \text{if } \dot{J}_s(x) = (\nabla J_s(x))^T \\ & \times (f(x) + g(x)\hat{u} + k(x)\hat{v}) < 0, \\ 1, & \text{else.} \end{cases}\quad (37)$$

It should be mentioned that the first term in the tuning law (36) seeks to minimise (35) and the remainder of (36) ensures the system states remain bounded while the neural network approximator learns the optimal cost function.

The process of deriving the additional terms is as follows:

Denote the derivative of the Lyapunov function candidate for system (2) with control pair (8) as

$$\begin{aligned}\Phi = & (\nabla J_s(x))^T \left(f(x) - \frac{1}{2} g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T \hat{w}_c \right. \\ & \left. + \frac{1}{2} k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T \hat{w}_c \right)\end{aligned}\quad (38)$$

If the closed-loop system is unstable, we can obtain $\Phi > 0$. In order to keep the closed-loop system stable, i.e. $\Phi < 0$, by using the gradient descent method, we have

$$\begin{aligned}-\frac{\partial \Phi}{\partial \hat{w}_c} = & \frac{1}{2} \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) \nabla J_s(x) \\ & - \frac{1}{2} \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) \nabla J_s(x)\end{aligned}\quad (39)$$

Thus, the additional term (39) is necessary for the proposed tuning law to ensure the stability of the closed-loop system during the online learning process.

Definition (37) shows that when the nonlinear system is stable, the second and the third terms of (36) are removed. In contrast, when the system exhibits signs of instability, the second and the third terms of (36) are activated for learning process (Dierks & Jagannathan, 2010).

In order to prove the stability of above tuning law, the dynamics of the critic weight estimation error should be given. Considering $\tilde{w}_c = w_c - \hat{w}_c$ and

$$\begin{aligned}\frac{\partial e_c}{\partial \hat{w}_c} = & \nabla \sigma_c(x) f(x) \\ & - \frac{1}{2} \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T \hat{w}_c \\ & + \frac{1}{2} \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T \hat{w}_c\end{aligned}\quad (40)$$

the error dynamics is obtained

$$\begin{aligned}\dot{\hat{w}}_c = & \alpha_c \left\{ -\tilde{w}_c^T \nabla \sigma_c(x) f(x) \right. \\ & - \frac{1}{4} \tilde{w}_c^T \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T \tilde{w}_c \\ & + \frac{1}{2} \tilde{w}_c^T \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T w_c \\ & + \frac{1}{4} \tilde{w}_c^T \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T \tilde{w}_c \\ & - \frac{1}{2} \tilde{w}_c^T \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T w_c - \varepsilon_H \left. \right\} \\ & \cdot \left\{ \nabla \sigma_c(x) f(x) - \frac{1}{2} \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T w_c \right. \\ & + \frac{1}{2} \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T \tilde{w}_c \\ & + \frac{1}{2} \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T w_c \\ & - \frac{1}{2} \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T \tilde{w}_c \left. \right\} \\ & - \frac{1}{2} \alpha_s \sum (x, \hat{u}(t), \hat{v}(t)) \tilde{w}_c^T \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) \nabla J_s(x) \\ & + \frac{1}{2} \alpha_s \sum (x, \hat{u}(t), \hat{v}(t)) \tilde{w}_c^T \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) \nabla J_s(x)\end{aligned}\quad (41)$$

Assumption 4.1: Without loss of generality,

- (1) The internal dynamics $f(x)$ is Lipschitz, the input dynamics $g(x)$ and $k(x)$ are bounded, i.e. there exist constants g_m , g_M ($0 < g_m < g_M$) and k_m , k_M ($0 < k_m < k_M$) such that

$$g_m < \|g(x)\| < g_M \text{ and } k_m < \|k(x)\| < k_M.$$

- (2) The ideal neural network weight w_c is bounded by a known constant w_M , i.e. $\|w_c\| \leq w_M$.
- (3) The neural network approximation error $\varepsilon_c(x)$ and its gradient $\nabla \varepsilon_c(x)$ are bounded such that $\|\varepsilon_c(x)\| < b_\varepsilon$ and $\|\nabla \varepsilon_c(x)\| < b_{\varepsilon_x}$.
- (4) The neural network activation function $\sigma_c(x)$ and its gradient $\nabla \sigma_c(x)$ are also bounded such that $\|\sigma_c(x)\| < b_\varphi$ and $\|\nabla \sigma_c(x)\| < b_{\varphi_x}$.

Definition 4.1 (Vamvoudakis et al., 2011): A time signal $\zeta(t)$ is said to be uniformly ultimately bounded (UUB) if there exists a compact set $S \subset \mathbb{R}^n$ so that for all $\zeta(0) \in S$ there exists a bound M and a time $T(M, \zeta(0))$ such that $\|\zeta(t)\| \leq M$ for all $t \geq t_0 + T$.

Theorem 4.1: For system (2), let control pair $\hat{u}(t), \hat{v}(t)$ be provided by (30). Let tuning law for the critic neural network be provided by (36). Then, the weight estimation error $\tilde{w}_c$ of the critic network is UUB.

Proof: Choose the Lyapunov function candidate as

$$L = \frac{1}{2\alpha_c} \tilde{w}_c^T \tilde{w}_c + \frac{\alpha_s}{2\alpha_c} J_s(x) \quad (42)$$

Take the first derivative with respect to time as

$$\dot{L} = \frac{1}{\alpha_c} \tilde{w}_c^T \dot{\tilde{w}}_c + \frac{\alpha_s}{2\alpha_c} (\nabla J_s(x))^T \dot{x} \quad (43)$$

Substituting (41) into (43) yields

$$\begin{aligned} \dot{L} = & \tilde{w}_c^T \left\{ -\tilde{w}_c^T \nabla \sigma_c(x) f(x) \right. \\ & - \frac{1}{4} \tilde{w}_c^T \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T \tilde{w}_c \\ & + \frac{1}{2} \tilde{w}_c^T \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T w_c \\ & + \frac{1}{4} \tilde{w}_c^T \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T \tilde{w}_c \\ & \left. - \frac{1}{2} \tilde{w}_c^T \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T w_c - \varepsilon_H \right\} \\ & \times \left\{ \nabla \sigma_c(x) f(x) - \frac{1}{2} \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T w_c \right. \\ & + \frac{1}{2} \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T \tilde{w}_c \\ & + \frac{1}{2} \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T w_c \\ & \left. - \frac{1}{2} \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T \tilde{w}_c \right\} \\ & + \frac{\alpha_s}{2\alpha_c} (\nabla J_s(x))^T \dot{x} - \frac{\alpha_s}{2\alpha_c} \sum (x, \hat{u}(t), \hat{v}(t)) \\ & \tilde{w}_c^T \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) \nabla J_s(x) + \frac{\alpha_s}{2\alpha_c} \sum (x, \hat{u}(t), \hat{v}(t)) \\ & \tilde{w}_c^T \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) \nabla J_s(x) \end{aligned} \quad (44)$$

For simplicity, denote $A = \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T$, $B = \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T$. Then, formula

(44) becomes

$$\begin{aligned} \dot{L} = & \left(-\tilde{w}_c^T \nabla \sigma_c(x) f(x) - \frac{1}{4} \tilde{w}_c^T A \tilde{w}_c + \frac{1}{2} \tilde{w}_c^T A w_c \right. \\ & \left. + \frac{1}{4} \tilde{w}_c^T B \tilde{w}_c - \frac{1}{2} \tilde{w}_c^T B w_c - \varepsilon_H \right) \\ & \cdot \left(\tilde{w}_c^T \nabla \sigma_c(x) f(x) - \frac{1}{2} \tilde{w}_c^T A w_c + \frac{1}{2} \tilde{w}_c^T A \tilde{w}_c \right. \\ & \left. + \frac{1}{2} \tilde{w}_c^T B w_c - \frac{1}{2} \tilde{w}_c^T B \tilde{w}_c \right) + \frac{\alpha_s}{2\alpha_c} (\nabla J_s(x))^T \dot{x} \\ & - \frac{\alpha_s}{2\alpha_c} \sum (x, \hat{u}(t), \hat{v}(t)) \\ & \tilde{w}_c^T \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) \nabla J_s(x) + \frac{\alpha_s}{2\alpha_c} \sum (x, \hat{u}(t), \hat{v}(t)) \\ & \tilde{w}_c^T \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) \nabla J_s(x) \end{aligned} \quad (45)$$

Without loss of generality, assume $\lambda_{1m} > 0, \lambda_{2m} > 0$ and $\lambda_{1M} > 0, \lambda_{2M} > 0$ are the lower and upper bounds of the norm of matrices A and B , respectively, such that $\lambda_{1m} \leq \|A\| \leq \lambda_{1M}$ and $\lambda_{2m} \leq \|B\| \leq \lambda_{2M}$. Additionally, in sight of Assumption 4.1, it is reasonable to assume that $\|\nabla \sigma_c(x) f(x)\| \leq \lambda_3$, the norm $\|A w_c\| \leq \lambda_4$, $\|B w_c\| \leq \lambda_5$ and $\|\varepsilon_H\| \leq \lambda_6$, where $\lambda_3, \lambda_4, \lambda_5$ and λ_6 are both positive constants.

By taking norm and using the following inequalities

$$\begin{aligned} ab &= \frac{1}{2} \left(-\left(\phi_1 a - \frac{b}{\phi_1} \right)^2 + \phi_1^2 a^2 + \frac{b^2}{\phi_1^2} \right) \\ &\leq \frac{1}{2} \left(\phi_1^2 a^2 + \frac{b^2}{\phi_1^2} \right) \\ -ab &= -\frac{1}{2} \left(\left(\phi_2 a - \frac{b}{\phi_2} \right)^2 + \phi_2^2 a^2 + \frac{b^2}{\phi_2^2} \right) \\ &\leq \frac{1}{2} \left(\phi_2^2 a^2 + \frac{b^2}{\phi_2^2} \right) \end{aligned} \quad (46)$$

where ϕ_1 and ϕ_2 are nonzero constants, and some simple mathematical transformation, formula (45) can be rewritten as

$$\begin{aligned} \dot{L} \leq & - \left\{ \left(\frac{1}{8} - \frac{7}{16} \phi_2^2 \right) \lambda_{1m}^2 + \left(\frac{1}{8} - \frac{1}{16} \phi_2^2 \right) \lambda_{2m}^2 \right. \\ & \left. - \frac{1}{4} \phi_1^2 \lambda_{1M}^2 - \frac{7}{16} \phi_1^2 \lambda_{2M}^2 - \frac{1}{4} \lambda_{1M} \lambda_{2M} \right\} \|\tilde{w}_c\|^4 \\ & + \left\{ \left(\frac{5}{2} + \frac{3}{8 \phi_1^2} + \frac{1}{7} 16 \phi_2^2 \right) \lambda_3^2 + \left(\frac{3}{2} + \frac{3}{16 \phi_1^2} \right) \lambda_4^2 \right. \\ & \left. + \left(\frac{5}{4} + \frac{3}{16 \phi_1^2} + \frac{3}{16 \phi_2^2} \right) \lambda_5^2 \right\} \end{aligned}$$

$$\begin{aligned}
& -\frac{1}{4}\lambda_3\lambda_{2m} - \frac{1}{2}\lambda_{1m}\lambda_6 + \frac{1}{2}\lambda_{2M}\lambda_6 \Big\} \|\tilde{w}_c\|^2 + \lambda_6^2 \\
& + \frac{\alpha_s}{2\alpha_c} (\nabla J_s(x))^T \dot{x} - \frac{\alpha_s}{2\alpha_c} \sum (x, \hat{u}(t), \hat{v}(t)) \\
& \tilde{w}_c^T \nabla \sigma_c(x) g(x) R_1^{-1} g^T(x) \nabla J_s(x) + \frac{\alpha_s}{2\alpha_c} \sum (x, \hat{u}(t), \hat{v}(t)) \\
& \tilde{w}_c^T \nabla \sigma_c(x) k(x) R_2^{-1} k^T(x) \nabla J_s(x)
\end{aligned} \tag{47}$$

Remark 4.2: The inequalities (46) are used to avoid the terms $\|\tilde{w}_c\|^3$ and $\|\tilde{w}_c\|$, and thus, inequality (47) can be solved easily later. The constants ϕ_1 and ϕ_2 are introduced to guarantee the coefficient of the term $\|\tilde{w}_c\|^4$ is negative, which is critical to the convergence of the closed-loop system. Furthermore, the result of (47) is not unique and it may have better methods to deal with this inequality.

Now, the cases of $\sum (x, \hat{u}(t), \hat{v}(t)) = 0$ and $\sum (x, \hat{u}(t), \hat{v}(t)) = 1$ will be considered.

Case 1: The term $\sum (x, \hat{u}(t), \hat{v}(t)) = 0$, since $(\nabla J_s(x))^T \dot{x} < 0$, there exists a positive constant λ_7 such that $0 < \lambda_7 \|\nabla J_s(x)\| \leq -(\nabla J_s(x))^T \dot{x}$ holds (Wang et al., 2014), then, inequality (47) becomes

$$\dot{L} \leq -\lambda_8 \|\tilde{w}_c\|^4 + \lambda_9 \|\tilde{w}_c\|^2 + \lambda_6^2 - \frac{\alpha_s}{2\alpha_c} \lambda_7 \|\nabla J_s(x)\| \tag{48}$$

where

$$\begin{aligned}
\lambda_8 &= \left(\frac{1}{8} - \frac{7}{16} \phi_2^2 \right) \lambda_{1m}^2 + \left(\frac{1}{8} - \frac{1}{16} \phi_2^2 \right) \lambda_{2m}^2 \\
&\quad - \frac{1}{4} \phi_1^2 \lambda_{1M}^2 - \frac{7}{16} \phi_1^2 \lambda_{2M}^2 - \frac{1}{4} \lambda_{1M} \lambda_{2M} \tag{49} \\
\lambda_9 &= \left(\frac{5}{2} + \frac{3}{8\phi_1^2} + \frac{7}{16\phi_2^2} \right) \lambda_3^2 + \left(\frac{3}{2} + \frac{3}{16\phi_1^2} \right) \lambda_4^2 \\
&\quad + \left(\frac{5}{4} + \frac{3}{16\phi_1^2} + \frac{3}{16\phi_2^2} \right) \lambda_5^2 \\
&\quad - \frac{1}{4} \lambda_3 \lambda_{2m} - \frac{1}{2} \lambda_{1m} \lambda_6 + \frac{1}{2} \lambda_{2M} \lambda_6 \tag{50}
\end{aligned}$$

Therefore, whenever the inequality

$$\|\tilde{w}_c\| \geq \sqrt{\frac{\lambda_9 + \sqrt{\lambda_9^2 + 4\lambda_8\lambda_6^2}}{2\lambda_8}} \triangleq A_1 \tag{51}$$

holds, we have $\dot{L} < 0$.

Case 2: The term $\sum (x, \hat{u}(t), \hat{v}(t)) = 1$, which implies the learning process-based input (30) may not be stable. To complete the proof, add and subtract the following two

terms to (47):

$$\begin{aligned}
& \frac{\alpha_s}{2\alpha_c} (\nabla J_s(x))^T (g(x) R_1^{-1} g^T(x) (\nabla \sigma_c(x))^T w_c + \nabla \varepsilon_c(x)) \\
& \frac{\alpha_s}{2\alpha_c} (\nabla J_s(x))^T (k(x) R_2^{-1} k^T(x) (\nabla \sigma_c(x))^T w_c + \nabla \varepsilon_c(x))
\end{aligned} \tag{52}$$

and take some mathematical transformation yields

$$\begin{aligned}
\dot{L} &\leq -\lambda_8 \|\tilde{w}_c\|^4 + \lambda_9 \|\tilde{w}_c\|^2 + \lambda_6^2 \\
&\quad + \frac{\alpha_s}{2\alpha_c} (\nabla J_s(x))^T (f(x) + g(x)u^*(t) + k(x)v^*(t)) \\
&\quad + \frac{\alpha_s}{2\alpha_c} (\nabla J_s(x))^T D_1 \nabla \varepsilon_c(x) - \frac{\alpha_s}{2\alpha_c} (\nabla J_s(x))^T D_2 \nabla \varepsilon_c(x)
\end{aligned} \tag{53}$$

where $D_1 = g(x) R_1^{-1} g^T(x) > 0$ and $D_2 = k(x) R_2^{-1} k^T(x) > 0$ are both bounded by known constants

D_{1m}, D_{2m} and D_{1M}, D_{2M} , such that $D_{1m} \leq \|D_1\| \leq D_{1M}$ and $D_{2m} \leq \|D_2\| \leq D_{2M}$.

Recalling Assumption 3.1 such that $\lambda_{\min} \|\nabla J_s(x)\|^2 \leq (\nabla J_s(x))^T \Lambda(x) \nabla J_s(x) \leq \lambda_{\max} \|\nabla J_s(x)\|^2$, and Assumption 4.1, we obtain

$$\begin{aligned}
\dot{L} &\leq -\lambda_8 \|\tilde{w}_c\|^4 + \lambda_9 \|\tilde{w}_c\|^2 + \lambda_6^2 - \lambda_{10} \|\nabla J_s(x)\|^2 \\
&\quad + \lambda_{11} \|\nabla J_s(x)\|
\end{aligned} \tag{54}$$

where $\lambda_{10} = \frac{\alpha_s}{2\alpha_c} \lambda_{\max}$, $\lambda_{11} = \frac{\alpha_s}{2\alpha_c} (D_{1M} + D_{2M}) b_{\varepsilon_x}$

Therefore, whenever the inequality

$$\|\tilde{w}_c\| \geq \sqrt{\frac{\lambda_9}{2\lambda_8} + \sqrt{\frac{\lambda_9^2}{4\lambda_8^2} + \frac{\lambda_6^2}{\lambda_8} + \frac{\lambda_{11}^2}{4\lambda_8\lambda_{10}}}} \triangleq A_2 \tag{55}$$

holds, we have $\dot{L} < 0$.

In sight of Case 1 and Case 2, if the inequality $\|\tilde{w}_c\| > \max(A_1, A_2)$ holds, then, $\dot{L} < 0$ is guaranteed. Thus, according to the standard Lyapunov extension theorem (Lewis, Jagannathan, & Yesildirak, 1998), the weight estimation error $\tilde{w}_c$ is UUB with the ultimate bound A_1 or A_2 .

This completes the proof.

Next, we develop a theorem to show rigorously that the closed-loop system is stable in the sense of UUB during the critic network learning process.

Theorem 4.2: Consider system (2) with associated HJI equation (9), let control pair $\hat{u}(t), \hat{v}(t)$ be provided by (30), let tuning law for the critic neural network be provided by (36). Then, the closed-loop system is guaranteed to be UUB with the ultimate bound Ω_x in the subsequent equation (70).

Proof: Consider the Lyapunov function candidate $J^*(x)$ given in (6). Taking the derivative of $J^*(x)$ along the system trajectory $\dot{x} = f(x) + g(x)\hat{u}(t) + k(x)\hat{v}(t)$, we get

$$\begin{aligned} \dot{J}^*(x) &= (\nabla J^*(x))^T f(x) + (\nabla J^*(x))^T g(x)\hat{u}(t) \\ &\quad + (\nabla J^*(x))^T k(x)\hat{v}(t) \end{aligned} \quad (56)$$

From (8) and (9), we have

$$\begin{aligned} (\nabla J^*(x))^T f(x) &= -Q(x) - (\nabla J^*(x))^T g(x)u^*(t) \\ &\quad - (\nabla J^*(x))^T k(x)v^*(t) \\ &\quad - u^{*T}(t)R_1u^*(t) + v^{*T}(t)R_2v^*(t) \end{aligned} \quad (57)$$

Substituting (57) into (56) yields

$$\begin{aligned} \dot{J}^*(x) &= (\nabla J^*(x))^T g(x)(\hat{u}(t) - u^*(t)) \\ &\quad + (\nabla J^*(x))^T k(x)(\hat{v}(t) - v^*(t)) \\ &\quad - Q(x) - u^{*T}(t)R_1u^*(t) + v^{*T}(t)R_2v^*(t) \\ &\leq (\nabla J^*(x))^T g(x)(\hat{u}(t) - u^*(t)) \\ &\quad + (\nabla J^*(x))^T k(x)(\hat{v}(t) - v^*(t)) \\ &\quad - x^T Q_1 x - u^{*T}(t)R_1u^*(t) + v^{*T}(t)R_2v^*(t) \end{aligned} \quad (58)$$

Considering (26), we have

$$\begin{aligned} u^{*T}(t)R_1u^*(t) &= \frac{1}{4}w_c^T A w_c + \frac{1}{2}w_c^T \nabla \sigma_c(x) D_1 \nabla \varepsilon_c(x) \\ &\quad + \frac{1}{4}(\nabla \varepsilon_c(x))^T D_1 \nabla \varepsilon_c(x) \end{aligned} \quad (59)$$

$$\begin{aligned} v^{*T}(t)R_2v^*(t) &= \frac{1}{4}w_c^T B w_c + \frac{1}{2}w_c^T \nabla \sigma_c(x) D_2 \nabla \varepsilon_c(x) \\ &\quad + \frac{1}{4}(\nabla \varepsilon_c(x))^T D_2 \nabla \varepsilon_c(x) \end{aligned} \quad (60)$$

Based on the assumptions in this paper, we get

$$\begin{aligned} \|u^{*T}(t)R_1u^*(t)\| &\leq \frac{1}{4}\lambda_{1M}w_M^2 + \frac{1}{2}w_M D_{1M} b_{\phi_x} b_{\varepsilon_x} \\ &\quad + \frac{1}{4}D_{1M} b_{\varepsilon_x}^2 \end{aligned} \quad (61)$$

$$\begin{aligned} \|v^{*T}(t)R_2v^*(t)\| &\leq \frac{1}{4}\lambda_{2M}w_M^2 + \frac{1}{2}w_M D_{2M} b_{\phi_x} b_{\varepsilon_x} \\ &\quad + \frac{1}{4}D_{2M} b_{\varepsilon_x}^2 \end{aligned} \quad (62)$$

On the other hand, by using (26) and (30), we get

$$\begin{aligned} \hat{u}(t) - u^*(t) &= -\frac{1}{2}R_1^{-1}g^T(x)(\nabla \sigma_c(x))^T \tilde{w}_c \\ &\quad + \frac{1}{2}R_1^{-1}g^T(x)\nabla \varepsilon_c(x) \end{aligned} \quad (63)$$

$$\begin{aligned} \hat{v}(t) - v^*(t) &= \frac{1}{2}R_2^{-1}k^T(x)(\nabla \sigma_c(x))^T \tilde{w}_c \\ &\quad - \frac{1}{2}R_2^{-1}k^T(x)\nabla \varepsilon_c(x) \end{aligned} \quad (64)$$

Then, taking the norm of (63) and (64) yields

$$\|\hat{u}(t) - u^*(t)\| \leq \frac{1}{2}R_1^{-1}g_M b_{\phi_x} \|\tilde{w}_c\| + \frac{1}{2}R_1^{-1}g_M b_{\varepsilon_x} \quad (65)$$

$$\|\hat{v}(t) - v^*(t)\| \leq \frac{1}{2}R_2^{-1}k_M b_{\phi_x} \|\tilde{w}_c\| + \frac{1}{2}R_2^{-1}k_M b_{\varepsilon_x} \quad (66)$$

By Theorem 4.1, we know that $\tilde{w}_c$ is UUB with ultimate bound A_1 or A_2 . Let $\Gamma = \max(A_1, A_2)$. From formulas (65) and (66), we have

$$\|\hat{u}(t) - u^*(t)\| \leq \frac{1}{2}R_1^{-1}g_M b_{\phi_x} \Gamma + \frac{1}{2}R_1^{-1}g_M b_{\varepsilon_x} \quad (67)$$

$$\|\hat{v}(t) - v^*(t)\| \leq \frac{1}{2}R_2^{-1}k_M b_{\phi_x} \Gamma + \frac{1}{2}R_2^{-1}k_M b_{\varepsilon_x} \quad (68)$$

Hence, by using (61), (62), (67) and (68), (57) becomes

$$\dot{J}^*(x) \leq -\lambda_{\min}(Q_1)\|x\|^2 + \Lambda \quad (69)$$

where $\lambda_{\min}(Q_1)$ denotes the minimum eigenvalue of the matrix Q_1 and

$$\begin{aligned} \Lambda &= \frac{1}{2}(\Gamma + 1)(R_1^{-1}(g_M b_{\phi_x})^2 + R_2^{-1}(k_M b_{\phi_x})^2) \\ &\quad + \frac{1}{4}w_M^2(\lambda_{1M} + \lambda_{2M}) \\ &\quad + \frac{1}{2}w_M(D_{1M} + D_{2M})b_{\phi_x}b_{\varepsilon_x} + \frac{1}{4}(D_{1M} + D_{2M})b_{\varepsilon_x}^2 \end{aligned}$$

Therefore, (69) yields $\dot{J}^*(x) < 0$ as long as x is out of

$$\Omega_x = \left\{ x : \|x\| \leq \sqrt{\frac{\Lambda}{\lambda_{\min}(Q_1)}} \right\} \quad (70)$$

According to the standard Lyapunov extension theorem (Lewis et al., 1998), this verifies that the trajectory of the closed-loop system is UUB. That is, the closed-loop system is stable in the sense of UUB during the critic network learning process.

Remark 4.3: Noticing expression (70), we find that Ω_x can be kept small by selecting a large enough $\lambda_{\min}(Q_1)$. In addition, if the number of neurons N goes to infinity, there exists $\varepsilon_c(x) \rightarrow 0$ and $\nabla \varepsilon_c(x) \rightarrow 0$. Hence, b_{ε_x} can be arbitrarily small when N is large enough. Thus, the term Λ can be kept small.

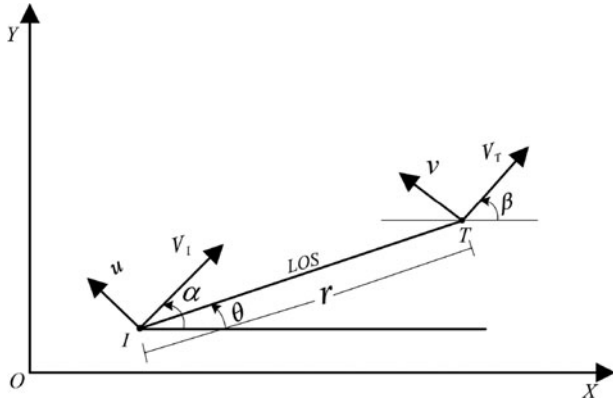


Figure 1. Engagement geometry.

5. Application to interceptor–target engagement guidance problem

In this section, we built the mathematical model of interceptor–target engagement scenario, first. Then, the guidance principle is described to ensure capture successfully. Finally, the uncertain nonlinear differential game framework is formulated to fit the proposed guidance law in Section 3 in interceptor–target engagement problem.

5.1 Engagement scenario

Consider a planar engagement scenario between an interceptor and a target as shown in Figure 1, V_I and V_T are velocity vectors of the interceptor and the target; their flight path angles (FPA) are α and β ; u and v are the control vectors perpendicular to the velocity vectors of the interceptor and the target, respectively. The interceptor and the target are assumed to have constant velocities. The line of sight (LOS) angle is given by θ , and LOS angular rate, $\dot{\theta}$, is designated by σ . The relative distance between the target and the interceptor, along the LOS, is given by r . The initial values of the variable quantities will be denoted with a subscript 0. The kinematics of the interceptor–target engagement system is given by the following equations:

$$V_r = \dot{r} = -V_T \cos(\beta - \theta) - V_I \cos(\alpha - \theta) \quad (71)$$

$$\sigma = \dot{\theta} = (V_T \sin(\beta - \theta) - V_I \sin(\alpha - \theta)) / r \quad (72)$$

$$\dot{\alpha} = u / V_I \quad (73)$$

$$\dot{\beta} = v / V_T \quad (74)$$

5.2 Guidance principle

To ensure capture, the interceptor has to select its strategy u to minimise the zero terminal miss distance while

the smart target will select its strategy v to maximise the terminal miss distance. Hence, the problem is posed in a form where the interceptor tries to minimise terminal miss distance, whereas the target tries to maximise it. Thus, ZEMD is defined as follows (Bardhan & Ghose, 2015):

Definition 5.1 (Zero effort miss distance): The quantity ZEMD at an instant t is defined as the closest distance between the interceptor and the target that would result if both the interceptor and the target do not maneuver from that instant onwards. The value of ZEMD, $r_{miss}(t)$, comes out to be

$$r_{miss}(t) = \frac{r^2 \sigma}{\sqrt{V_r^2 + r^2 \sigma^2}} \quad (75)$$

From the expression of $r_{miss}(t)$ in Equation (75), it can be seen that regulating σ to zero implies regulation of $r_{miss}(t)$ to zero value, and regulation of σ to zero while keeping $V_r < 0$ leads to interception.

5.3 Uncertain nonlinear differential game framework

According to the guidance principle, zeroing the LOS angular rate σ will lead to interception, then, the guidance law can be obtained by achieving the convergence of σ by adjusting control vector u . Select the system state as σ , i.e. $x = \sigma$, then, differentiating Equation (72) with respect to time yields

$$\begin{aligned} \dot{x} &= f(x) + g(x)u + k(x)v \\ &= -\frac{2V_r}{r}x - \frac{\cos(\alpha - \theta)}{r}u + \frac{\cos(\beta - \theta)}{r}v \end{aligned} \quad (76)$$

Note that, as u and v is multiplied by the term $\cos(\alpha - \theta)$ and $\cos(\beta - \theta)$, respectively, the LOS angular rate σ can be controlled for all $|(\alpha - \theta)| \neq \pi/2$, $|(\beta - \theta)| \neq \pi/2$. It can be shown that $|(\alpha - \theta)| = \pi/2$, $|(\beta - \theta)| = \pi/2$ is an unstable equilibrium.

Hence, the domain where the differential game-based guidance law is applicable is given by

$$\begin{aligned} \Phi = \{ &x : |(\alpha - \theta)| \neq \pi/2, |(\beta - \theta)| \neq \pi/2, r \neq 0, V_r < 0 \} \end{aligned} \quad (77)$$

Considering the model uncertainties of both the interceptor and the target, the interceptor–target engagement

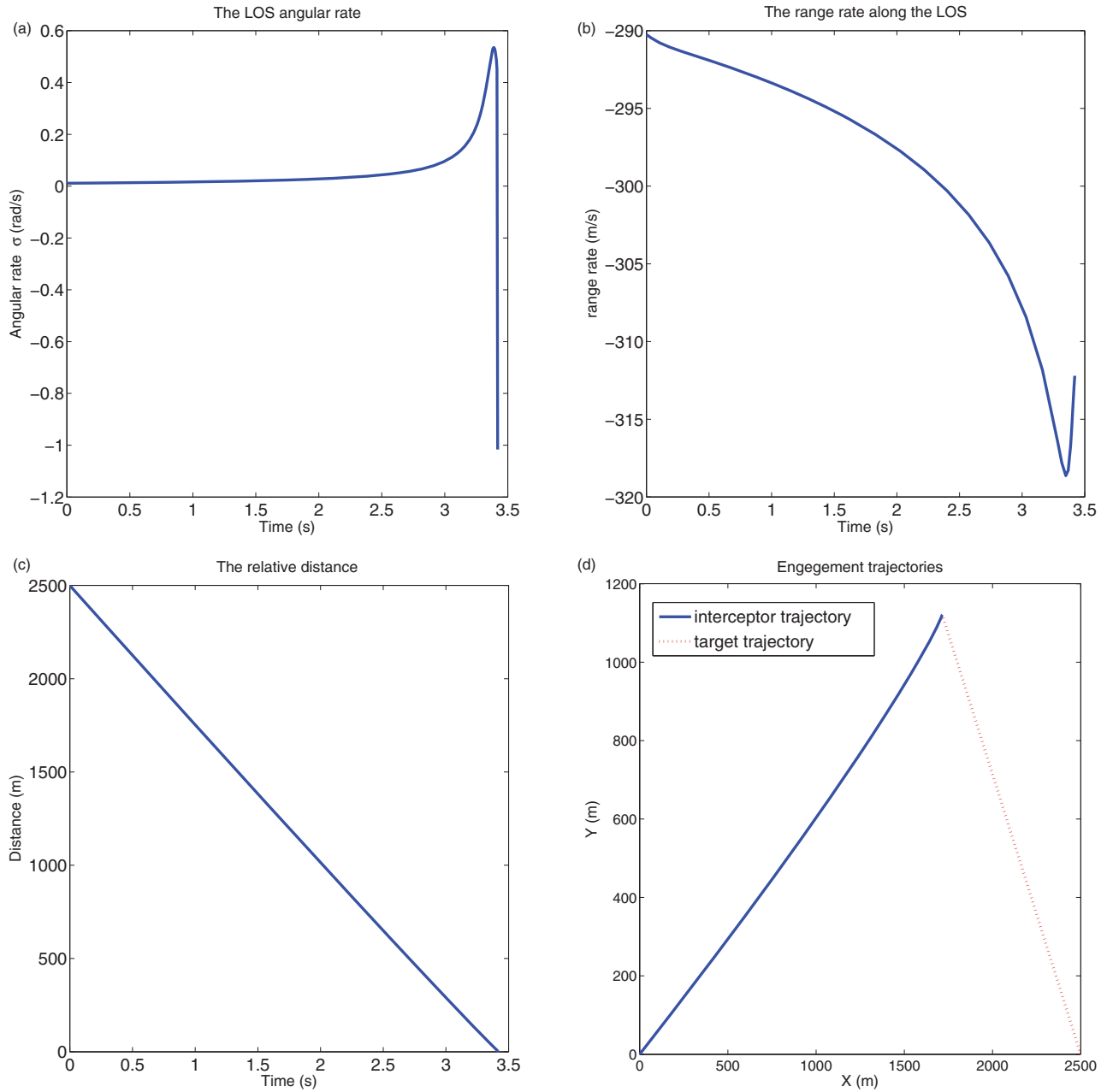


Figure 2. (a) The trajectory of the LOS angular rate. (b) The trajectory of the range rate. (c) The trajectory of relative distance. (d) The trajectory of engagement.

dynamics is described as

$$\begin{aligned}
 \dot{x} &= f(x) + \left(g(x) + \bar{d}(x)\right) u + \left(k(x) + \bar{b}(x)\right) v \\
 &= -\frac{2V_r}{r}x + \left(-\frac{\cos(\alpha - \theta)}{r} + \bar{d}(x)\right) u \\
 &\quad + \left(\frac{\cos(\beta - \theta)}{r} + \bar{b}(x)\right) v
 \end{aligned} \quad (78)$$

6. Simulations

In this section, the performance of guidance law derived from the differential games based on ADP/RL would be investigated using the obtained interceptor–target engagement models. The guidance law is derived assuming ideal autopilots for both the interceptor and the target; however, autopilots with first-order dynamics are considered for both the interceptor and the target for simulation study.

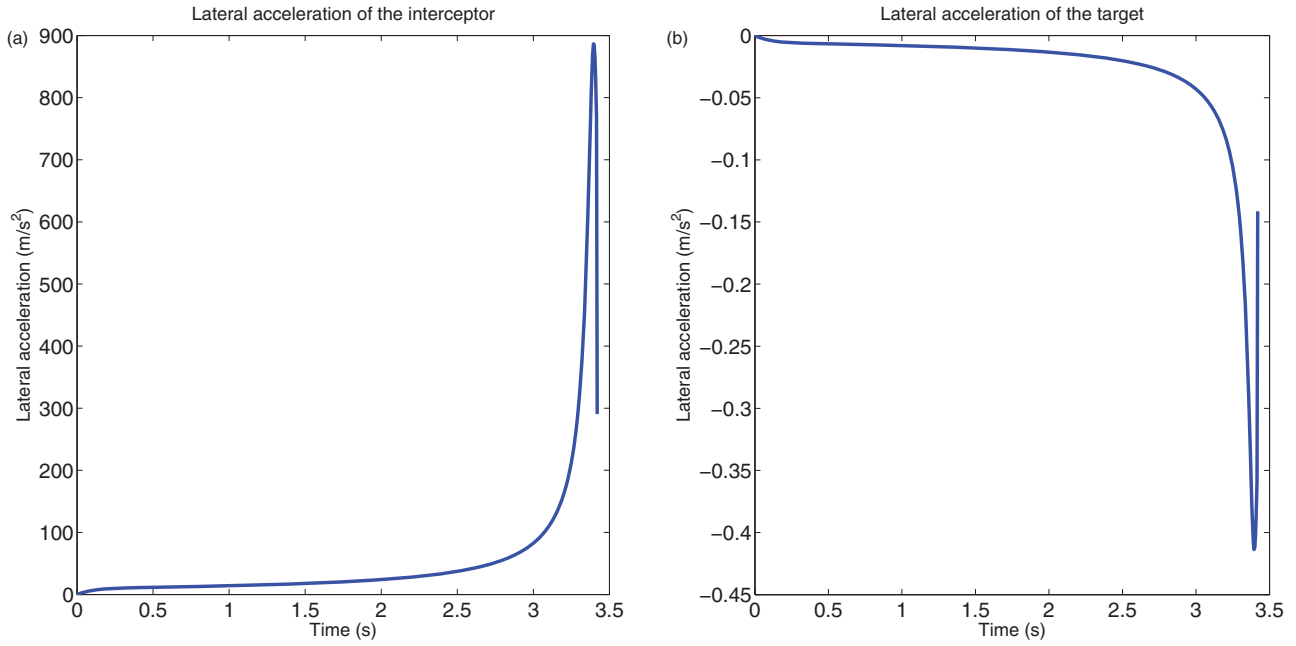


Figure 3. (a) The lateral acceleration of the interceptor. (b) The lateral acceleration of the target.

The equations of motion of the interceptor are as follows:

$$\begin{aligned}\dot{x}_I &= V_I \cos \alpha \\ \dot{y}_I &= V_I \sin \alpha \\ \dot{\alpha} &= a_I / V_I \\ \dot{a}_I &= (u - a_I) / \tau_I\end{aligned}\quad (79)$$

where (x_I, y_I) is the (x, y) coordinates of the interceptor, a_I denotes the lateral acceleration of the interceptor, τ_I is the time constant of the interceptor autopilots.

The equations of motion of the target are as follows:

$$\begin{aligned}\dot{x}_T &= V_T \cos \beta \\ \dot{y}_T &= V_T \sin \beta \\ \dot{\beta} &= a_T / V_T \\ \dot{a}_T &= (v - a_T) / \tau_T\end{aligned}\quad (80)$$

where (x_T, y_T) is the (x, y) coordinates of the target, a_T denotes the lateral acceleration of the target, τ_T is the time constant of the target autopilots.

The interceptor and the target velocity are considered to be constant through the engagement period and are taken to be $V_I = 600$ m/s and $V_T = 400$ m/s. The time constants of the interceptor and the target autopilots are $\tau_I = 0.1$ s and $\tau_T = 0.1$ s, respectively.

Assume the initial FPA angles of the interceptor and the target are $\alpha_0 = 30^\circ$ and $\beta_0 = 125^\circ$, the initial LOS angle $\theta_0 = 0^\circ$, the initial coordinates of the interceptor and the target are $(x_I, y_I) = (0, 0)$ and $(x_T, y_T) = (2500, 0)$. For the robust differential games guidance

laws, the constant gains are $\varepsilon_1 = 115$ and $\varepsilon_2 = 10$. The constant scalar $c = 36$ and the weighted matrices $Q_1 = 200$, $R_1 = 0.1$ and $R_2 = 0.1$ are included in the Hamiltonian function. The initial weights of the critic network are $w_{c0} = [50 \ 50 \ 50 \ 50]^T$ and the activation function is $\sigma_c = [x \ x^2 \ x^3 \ x^4]^T$, the learning rate is $\alpha_c = 0.07$ and $\alpha_s = 10$, respectively, the Lyapunov function candidate is $J_s(x) = \frac{1}{2}x^2$.

Assume that the term $\bar{d}(x) = \delta_1 x \cos(\delta_2 x + \delta_3 x^2)$ reflects the uncertainty of the interceptor, and the term $\bar{b}(x) = \delta_4 x \sin(x)$ reflects the uncertainty of the target, where δ_1 , δ_2 , δ_3 and δ_4 are unknown parameters with $\delta_1 \in [-10, 10]$, $\delta_2 \in [-25, 25]$, $\delta_3 \in [-15, 15]$ and $\delta_4 \in [-1, 1]$.

After simulation, the state variable $x = \sigma$ is regulated to zero at about 3.4 s, as shown in Figure 2(a), which leads the ZEMD to zero according to Equation (75). Meanwhile, Figure 2(b) shows that a negative V_r is maintained to ensure that the relative distance between the interceptor and the target, r , shown in Figure 2(c), decreases. Thus, the miss-distance is less than 0.5 m, and an interception is achieved successfully in Figure 2(d) in the presence of the model uncertainties. Figure 3 shows the corresponding lateral acceleration profiles of the interceptor and the target, in which the lateral acceleration shoots up toward the end of the engagement as r decreases sharply. In addition, the weights of the critic network and the residual error are shown in Figure 4, which shows that the critic network is effective to implement the ADP approach for finding the Nash solution of the nonlinear

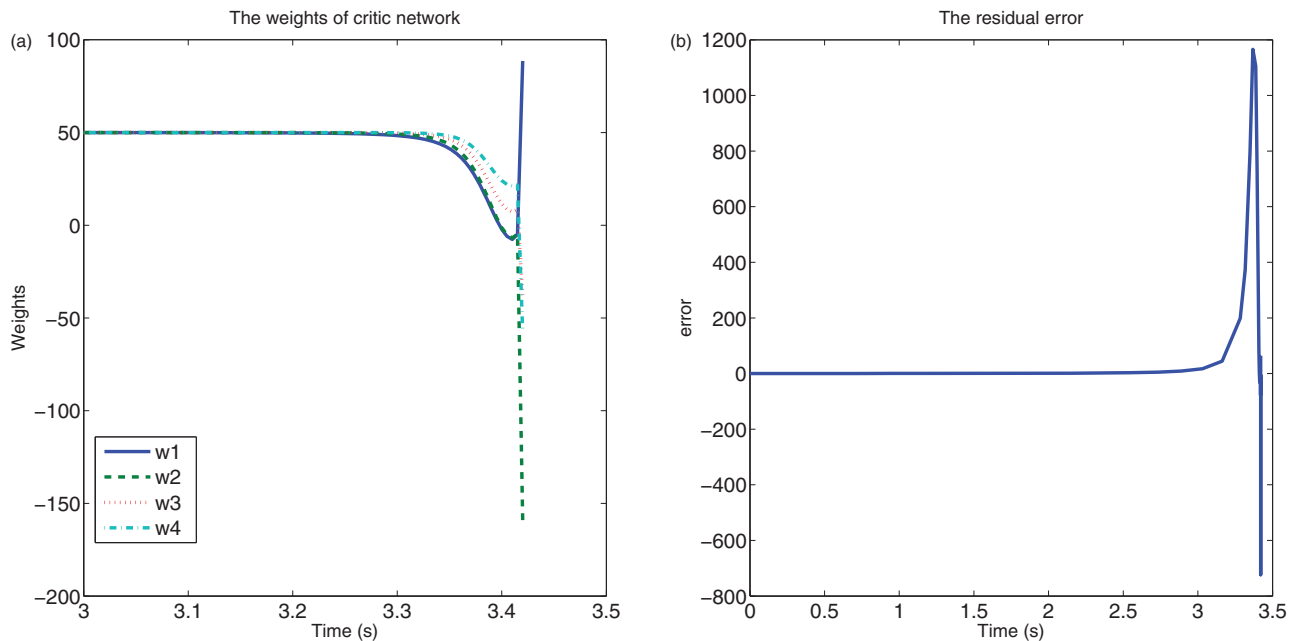


Figure 4. (a) The weights of the critic network. (b) The residual error.

two-player zero-sum differential games. Unlike the existing literatures related to ADP, the network weights do not converge to the optimal values, rather than shoot up at the end of the engagement as the residual error behaves erratically.

In fact, the erratic behaviours of the networks and the residual error in Figure 4 are coincident with the behaviours of the angular rate in Figure 2 and the lateral acceleration in Figure 3. A few seconds before interception, the residual error displays a sharp curve towards zero, and then seems to go quickly towards negative infinity. The reason for this behaviour is that the guidance law is based on the value of the relative distance r between the interceptor and the target. Thus, this sharp curve at the end of the engagement is expected, as the lateral acceleration of the two players increases as the relative distance r approaches zero, leading to larger errors in the LOS angular rate. In practice, once r gets small, it is best to stop calculating the lateral acceleration of the target and simply compute the lateral acceleration of the interceptor based upon the last value calculated for r .

The simulation results present the effectiveness of the proposed robust differential game guidance law based on ADP, in this paper.

7. Conclusion

In this paper, we have developed a novel robust differential game approach for a class of uncertain nonlinear two-player zero-sum differential game systems via the ADP technique. The robustness and optimality of the

control pair are proved under a specified cost function. The corresponding HJI equation is solved by constructing a critic network. The closed-loop system and the weight estimation error of critic network are proved to be stable in the sense of UUB. An interceptor–target engagement is considered by using the proposed robust differential game approach to verify its effectiveness and guidance performance is presented.

Disclosure statement

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